

SCHOOL OF COMPUTER SCIENCE AND IT

YBN UNIVERSITY RANCHI

JAVA PROGRAMMING

ASSIGNMENT 15

Problem 1: Rectangle Dimensions

Description:

Create a `Rectangle` class with attributes for `length` and `width`. Implement multiple constructors: one default constructor that initializes the rectangle with default values (e.g., `length = 1` and `width = 1`), and another constructor that initializes the rectangle with specific values. Implement methods to calculate the area and perimeter of the rectangle. Write a Java program to create `Rectangle` objects using both constructors and display their dimensions, area, and perimeter.

Concepts Covered:

- Default and parameterized constructors
- Using constructors to initialize object attributes
- Calculating and displaying results

Tasks:

1. Create a `Rectangle` class with a default constructor and a parameterized constructor.
2. Implement methods to calculate area and perimeter.
3. Write a `Main` class to create `Rectangle` objects using both constructors and display their dimensions, area, and perimeter.

Problem 2: Employee Records

Description:

Design an `Employee` class with attributes for `name`, `employeeId`, and `salary`. Implement multiple constructors: a default constructor that initializes the attributes to default values, and a parameterized constructor that initializes the attributes with specific values. Implement a method to display employee details. Write a Java program to create `Employee` objects using both constructors and display their details.

Concepts Covered:

- Default and parameterized constructors
- Initializing object attributes with constructor parameters
- Displaying object details

Tasks:

1. Create an `Employee` class with a default constructor and a parameterized constructor.
2. Implement a method to display employee details.

3. Write a `Main` class to create `Employee` objects using both constructors and display their details.

Problem 3: Book Library

Description:

Create a `Book` class with attributes for `title`, `author`, and `isbn`. Implement multiple constructors: a default constructor that initializes the attributes with default values, and a parameterized constructor that initializes the attributes with specific values. Implement a method to display book details. Write a Java program to create `Book` objects using both constructors and display their details.

Concepts Covered:

- Constructors with and without parameters
- Initializing and displaying object attributes
- Working with multiple constructors

Tasks:

1. Create a `Book` class with a default constructor and a parameterized constructor.
2. Implement a method to display book details.
3. Write a `Main` class to create `Book` objects using both constructors and display their details.

Problem 4: Circle with Radius

Description:

Design a `Circle` class with an attribute for `radius`. Implement a default constructor that initializes the radius to a default value (e.g., `radius = 1`) and a parameterized constructor that initializes the radius with a specific value. Implement methods to calculate the area and circumference of the circle. Write a Java program to create `Circle` objects using both constructors and display their radius, area, and circumference.

Concepts Covered:

- Default and parameterized constructors
- Calculating area and circumference using constructor parameters
- Displaying results

Tasks:

1. Create a `Circle` class with a default constructor and a parameterized constructor.
2. Implement methods to calculate area and circumference.
3. Write a `Main` class to create `Circle` objects using both constructors and display their radius, area, and circumference.

Problem 5: Student Profile

Description:

Create a `Student` class with attributes for `name`, `studentId`, and `gpa`. Implement a default constructor that initializes the attributes with default values, and a parameterized constructor that initializes the attributes with specific values. Implement methods to display student information and check if the student is on the honor roll (e.g., `gpa >= 3.5`). Write a Java program to create `Student` objects using both constructors, check their honor roll status, and display their information.

Concepts Covered:

- Default and parameterized constructors
- Constructor parameter usage
- Checking conditions and displaying results

Tasks:

1. Create a `Student` class with a default constructor and a parameterized constructor.
2. Implement methods to display student information and check honor roll status.
3. Write a `Main` class to create `Student` objects using both constructors, check their honor roll status, and display their information.